

The conductance step: closed-form integration

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The model. For postsynaptic neuron i , with $g_{ij} \geq 0$ the conductance of edge $j \rightarrow i$, E_{ij} the reversal that edge drives toward, E_i the leak reversal of the cell itself (flyvis stores it as the parameter `bias`, one learned value per cell type), stim_i the external stimulus current and τ_i the membrane time constant,

$$\tau_i \frac{dv_i}{dt} = -v_i + E_i + \sum_j g_{ij} u_j (E_{ij} - v_i) + \text{stim}_i, \quad u_j \equiv \text{relu}(v_j(t)). \quad (1)$$

u_j is the presynaptic drive. It is a *known number* at the moment the step is taken — the state of the other neurons at time t — not an unknown of the equation for v_i , which is what makes the closed form below available.

It is affine in v_i . Define the conductance-weighted sum of the reversals reaching i , and the total synaptic conductance onto i in units of its own leak conductance:

$$S_i = \sum_j g_{ij} u_j E_{ij}, \quad G_i = \sum_j g_{ij} u_j \quad (\geq 0). \quad (2)$$

Both are known at time t . The synaptic sum then splits into a part independent of v_i and a part proportional to it,

$$\sum_j g_{ij} u_j (E_{ij} - v_i) = \underbrace{\sum_j g_{ij} u_j E_{ij}}_{S_i} - v_i \underbrace{\sum_j g_{ij} u_j}_{G_i} = S_i - v_i G_i, \quad (3)$$

so the right-hand side of (1) collects the two v_i terms — the leak $-v_i$ and the shunt $-v_i G_i$ — into one:

$$\begin{aligned} \tau_i \frac{dv_i}{dt} &= -v_i + E_i + S_i - v_i G_i + \text{stim}_i \\ &= -(1 + G_i) v_i + (E_i + S_i + \text{stim}_i) \\ &= (1 + G_i) \left[\frac{E_i + S_i + \text{stim}_i}{1 + G_i} - v_i \right]. \end{aligned} \quad (4)$$

The bracket is the distance to the voltage at which the right-hand side vanishes, so (1) is a pure relaxation,

$$\tau_i \frac{dv_i}{dt} = (1 + G_i)(v_i^\infty - v_i), \quad v_i^\infty = \frac{E_i + S_i + \text{stim}_i}{1 + G_i}, \quad \tau_i^{\text{eff}} = \frac{\tau_i}{1 + G_i}. \quad (5)$$

Integration over one step. G_i , S_i and stim_i are read off the state at time t and held there across the step, as forward Euler also does. Then v_i^∞ and τ_i^{eff} are constants over $[t, t + \Delta t]$, (5) separates, and the integral is elementary:

$$\int_{v_i(t)}^{v_i(t+\Delta t)} \frac{dv_i}{v_i^\infty - v_i} = \int_t^{t+\Delta t} \frac{dt'}{\tau_i^{\text{eff}}} \implies \ln \frac{v_i^\infty - v_i(t)}{v_i^\infty - v_i(t + \Delta t)} = \frac{\Delta t}{\tau_i^{\text{eff}}} \equiv z, \quad (6)$$

so that, with $z = (1 + G_i)\Delta t/\tau_i$ the number of effective relaxation times one step covers,

$$\boxed{v_i(t + \Delta t) = v_i^\infty + (v_i(t) - v_i^\infty) e^{-z}} \quad (7)$$